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Unit “ 4 : Session “ 6 : SLO “ 2 | Parallel Computing “ Architecture Matching

■ I. Match the Following

1. Shared Memory “ Common address space: all processors read/write one global memory.
2. Distributed Memory “ Message passing: each node has private memory; data moves via explicit send/receive (MPI).
3. Multicore CPU “ Parallel execution units: several cores on one chip execute threads simultaneously.

■ II. Key Notes

- Flynn's taxonomy context: multicore CPUs and shared-memory machines are typically MIMD systems.
- Hybrid systems combine both: clusters of multicore nodes use shared memory inside a node and message passing across nodes.